

GOSFORD HIGH SCHOOL.
Extension 2 Mathematics.
HSC Assessment Task 1 December 2008.

Time Allowed: 2 hours.

Show all necessary working.

Start each section on a new page.

Section 1: Complex Numbers

Question 1.

(a) If $z = \sqrt{3} + i$ find

(i) $|z|$ (1)

(ii) $\arg(z)$ (1)

(iii) z^2 (2)

(iv) $\frac{1}{z}$ (2)

(v) iz (1)

(b) Plot $z, \bar{z}, z^2, \frac{1}{z}, iz$ on an Argand Diagram labelling them as P, Q, R, S, and T. (2)

Question 2.

(a) Express $z = 1 - \sqrt{3}i$ in mod-arg form. (2)

(b) Hence find z^6 in the form $a + ib$ where a and b are real. (2)

Question 3.

Sketch the region on an Argand Diagram for which

$$|z-1| \leq 4, \quad \frac{-\pi}{6} \leq \arg(z) \leq \frac{\pi}{6}, \quad z + \bar{z} \geq 6 \text{ hold simultaneously.} \quad (3)$$

Question 4.

The quadratic equation $z^2 + (1+i)z + k = 0$ has a root of $1 - 2i$. Find, in the form $a + ib$, the value of k and the other root of the equation. (3)

Question 5.

If $\omega = z^2 + \bar{z}^2$ find and draw a neat sketch of the locus of ω given that $\omega = 0$ (4)

Question 6.

(a) Show that for any two complex numbers z_1 & z_2 that:

$$\arg\left(\frac{z_1}{z_2}\right) = \arg(z_1) - \arg(z_2) \quad (2)$$

(b) Sketch the locus of z where $\arg\left(\frac{-z}{i}\right) = \arg\left(\frac{1}{z}\right)$. (3)

Question 7.

(a) Find the five fifth roots of 1 and indicate their position on an Argand Diagram. (4)

(b) If ω is one root of $z^5 = 1$, find the value of $\omega + \omega^2 + \omega^3 + \omega^4$. (2)

Question 8.

(a) Use DeMoivre's Theorem with $n = 2$ to show that:

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta \text{ and } \sin 2\theta = 2\sin \theta \cos \theta. \quad (2)$$

(b) Hence express $\cos \frac{2\pi}{n}$ & $\sin \frac{2\pi}{n}$ in terms of $\cos \frac{\pi}{n}$ & $\sin \frac{\pi}{n}$. (1)

(c) Using the result of (b) prove that:

$$\left(1 + \cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n}\right)^n = -2^n \cos^n \frac{\pi}{n}. \quad (4)$$

Question 9.

(a) Show that $\frac{\pi}{6} + \frac{\pi}{4} = \frac{5\pi}{12}$ and hence that $\tan \frac{5\pi}{12} = \frac{\sqrt{3} + 1}{\sqrt{3} - 1}$. (2)

Find a similar expression for $\tan \frac{\pi}{12}$ (2)

(b) Express $\sqrt{-6i}$ in the form $a+ib$ where a and b are real. (2)

(c) Solve $z^2 + (1+i)z + 2i = 0$ expressing the roots in the form $x+iy$ where x and y are real. (2)

(d) If these two roots are z_1 & z_2 prove that:

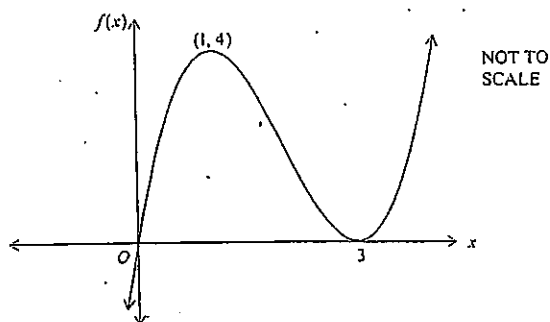
$$(i) \quad |z_1| = |z_2| = \sqrt{2}. \quad (1)$$

$$(ii) \quad \arg(z_1) + \arg(z_2) = \frac{\pi}{2}. \quad (2)$$

Section 2: Graphs (Start a new page.)

Question 1.

The function defined by $f(x) = x(x-3)^2$ is drawn below.



Draw separate, one-third page sketches, of each of the following:

(a) $y = |f(x)|$ (2)

(b) $y = f(|x|)$ (2)

(c) $y = f(-x)$ (2)

(d) $y = \frac{1}{f(x)}$ (2)

(e) $y^2 = f(x)$ (2)

Question 2.

Draw a neat sketch of $y = 1 + x^2$ and hence sketch on separate, one-third page diagrams, each of the following:

(a) $y = \frac{1}{1+x^2}$ (2)

(b) $y = \frac{1+x^2}{x}$ (2)

(c) $y = \left| \frac{1+x^2}{x} \right|$ (2)

(d) $y = \sqrt{\frac{1+x^2}{x}}$ (2)

Question 3.

(a) Sketch the graphs of $f(x) = x - 2$ and $g(x) = \frac{3}{x+2}$ on the same number plane. (2)

(b) Show that $\frac{x^2-1}{x+2} = x-2 + \frac{3}{x+2}$ (2)

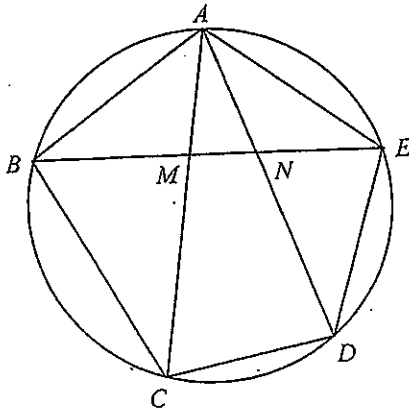
(c) By considering the sum of $f(x)$ & $g(x)$ sketch the graph of $y = \frac{x^2-1}{x+2}$ (2)

(d) Hence solve the inequality $\frac{x^2-1}{x+2} \leq 0$ (2)

Section 3: Circle Geometry (Start a new page.)

Question 1.

ABCDE is a pentagon inscribed in a circle. $AB = AE$. BE meets AC and AD at M and N respectively.



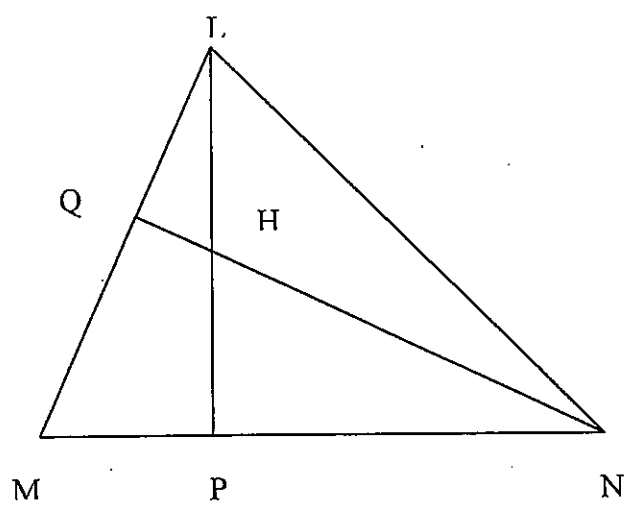
(a) Show that $\angle BEA = \angle ACE$. (2)

(b) Hence show CDMN is a cyclic quadrilateral. (3)

Question 2.

In an acute angled triangle with vertices L, M and N the foot of the perpendicular from L to MN is P and the foot of the perpendicular from N to LM is Q. The lines LP and QN intersect at H.

(P.T.O.)

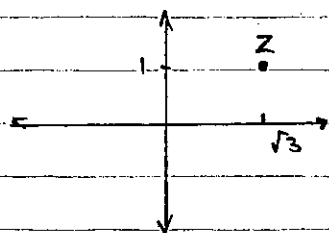


- (a) Prove that $\angle PHM = \angle PQM$. (2)
- (b) Prove that $\angle PHM = \angle LNM$. (2)
- (c) Produce MH to meet LN at R. Prove that $MR \perp LN$. (3)

SOLUTIONS.

COMPLEX NUMBERS:

Q1 a) $z = \sqrt{3} + i$



(i) $|z| = \sqrt{\sqrt{3}^2 + 1^2}$
 $= 2$

(1)

(ii) $\arg(z) = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$
 $= \frac{\pi}{6}$

(1)

(iii) $z^2 = (\sqrt{3} + i)^2$
 $= 3 + 2\sqrt{3}i + i^2$
 $= 2 + 2\sqrt{3}i$

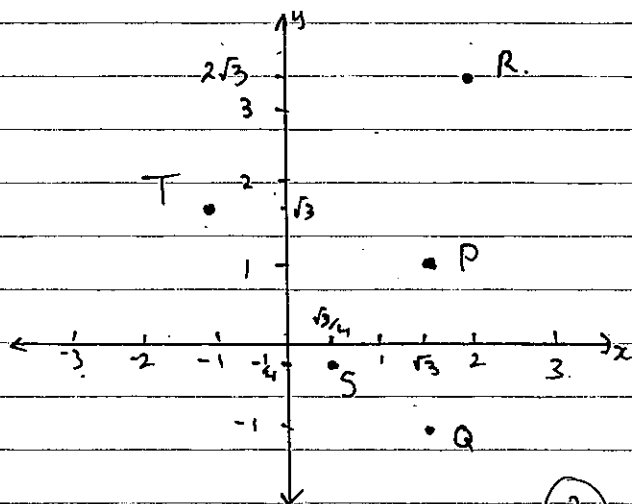
(2)

(iv) $\frac{1}{z} = \frac{1}{\sqrt{3} + i} \times \frac{\sqrt{3} - i}{\sqrt{3} - i}$
 $= \frac{\sqrt{3} - i}{4}$

(2)

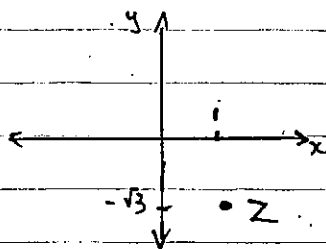
(v) $iz = i(\sqrt{3} + i)$
 $= \sqrt{3}i + i^2$
 $= -1 + \sqrt{3}i$

(1)



(2)

Q2 a) $z = 1 - \sqrt{3}i$



$|z| = 2$, $\arg(z) = -\frac{\pi}{3}$

$\therefore z = 2 \operatorname{cis}\left(-\frac{\pi}{3}\right)$

(2)

b) $z^6 = \left[2 \operatorname{cis}\left(-\frac{\pi}{3}\right)\right]^6$

$= 2^6 \operatorname{cis} -\frac{6\pi}{3}$

$= 64 \operatorname{cis} - 2\pi$

$= 64 \operatorname{cis} 0$

$= 64 (\cos 0 + i \sin 0)$

$= 64 (1 + 0i)$

$= 64 + 0i$

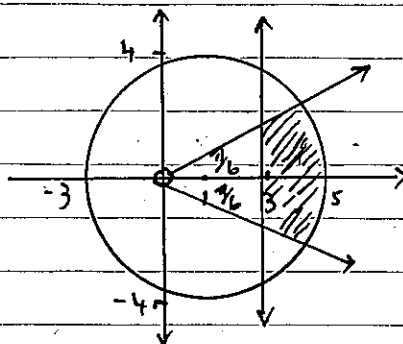
(2)

Q3. If $z + \bar{z} \geq 6$

$x + iy + x - iy \geq 6$

$2x \geq 6$

$x \geq 3$



(3)

Q4.

$$z^2 + (1+i)z + k = 0$$

When $z = 1-2i$

$$(1-2i)^2 + (1+i)(1-2i) + k = 0$$

$$-3+4i + 3-i + k = 0$$

$$-5i + k = 0$$

$$k = 5i. \quad (2)$$

If the roots are α, β

$$\alpha + \beta = -(1+i)$$

$$(1-2i) + \beta = -1-i$$

$$\beta = -2+i. \quad (1)$$

Q5. Let $z = x+iy$

$$w = (x+iy)^2 + (x-iy)^2$$

$$= x^2 - y^2 + 2xyi + x^2 - y^2 - 2xyi$$

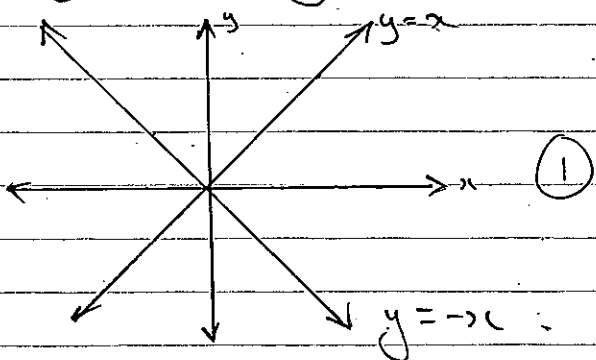
$$= 2x^2 - 2y^2$$

$$\therefore 2x^2 - 2y^2 = 0$$

$$x^2 - y^2 = 0$$

$$(x+y)(x-y) = 0$$

$$\therefore y = x \text{ or } y = -x.$$



Q6, a) Let $z_1 = r_1 (\cos \theta_1 + i \sin \theta_1)$

$$z_2 = r_2 (\cos \theta_2 + i \sin \theta_2)$$

$$\frac{z_1}{z_2} = \frac{r_1 (\cos \theta_1 + i \sin \theta_1)}{r_2 (\cos \theta_2 + i \sin \theta_2)}$$

$$= \frac{r_1}{r_2} \frac{[\cos \theta_1 + i \sin \theta_1]}{[\cos \theta_2 + i \sin \theta_2]}$$

$$= \frac{r_1}{r_2} \frac{[\cos \theta_1 + i \sin \theta_1]}{[\cos \theta_2 + i \sin \theta_2]} \times \frac{[\cos \theta_2 - i \sin \theta_2]}{[\cos \theta_2 - i \sin \theta_2]}$$

$$= \frac{r_1}{r_2} \frac{[\cos \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2 + i (\sin \theta_1 \cos \theta_2 - \cos \theta_1 \sin \theta_2)]}{\cos^2 \theta_2 + \sin^2 \theta_2}$$

$$= \frac{r_1}{r_2} [\cos (\theta_1 - \theta_2) + i \sin (\theta_1 - \theta_2)]$$

$$\therefore \arg \left(\frac{z_1}{z_2} \right) = \theta_1 - \theta_2$$

$$= \arg z_1 - \arg z_2 \quad (2)$$

b) If $\arg \left(\frac{-z}{i} \right) = \arg \left(\frac{1}{2} \right)$

$$\arg(-z) - \arg(i) = \arg(1) - \arg(2)$$

$$\arg(-1) + \arg(z) - \arg(i) = \arg(1) - \arg(2)$$

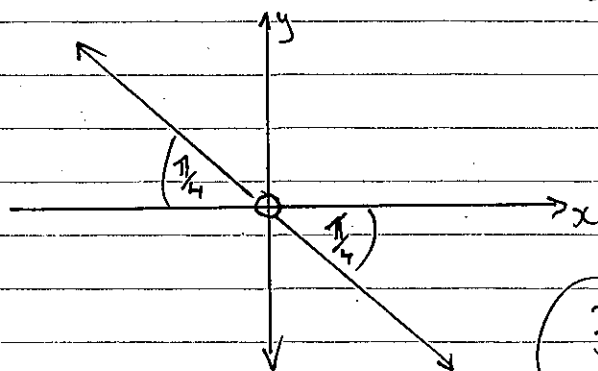
$$\arg(-1) + 2\arg(z) = 0 + \frac{\pi}{2}$$

$$\pm \pi + 2\arg(z) = \frac{\pi}{2}$$

$$2\arg(z) = \mp \pi + \frac{\pi}{2}$$

$$2\arg(z) = -\frac{\pi}{2} \text{ or } \frac{3\pi}{2}$$

$$\arg(z) = -\frac{\pi}{4} \text{ or } \frac{3\pi}{4}$$



Q7. a) $z^5 = 1$

Now $\cos(0+2k\pi) + i \sin(0+2k\pi)$

$R(\cos \phi + i \sin \phi) = \sqrt[5]{\cos(0+2k\pi) + i \sin(0+2k\pi)}$

$R^5(\cos 5\phi + i \sin 5\phi) = \cos(0+2k\pi) + i \sin(0+2k\pi)$

$\therefore R^5 = 1 \Rightarrow R = 1$

$5\phi = 0 + 2k\pi, k=0,1,2,3,4$

$\therefore$ the five 5th roots are

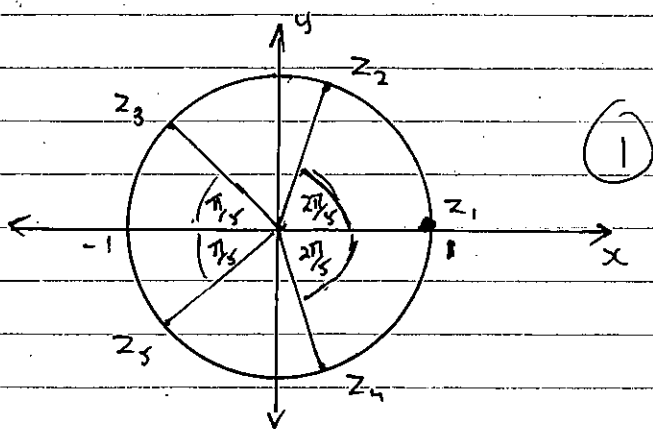
$\text{cis}\left(\frac{0+2k\pi}{5}\right) \quad k=0,1,2,3,4$

i.e. $1, \text{cis } \frac{2\pi}{5}, \text{cis } \frac{4\pi}{5}, \text{cis } \frac{6\pi}{5}, \text{cis } \frac{8\pi}{5}$

$\text{cis } \frac{8\pi}{5}$

(3)

or $1, \text{cis } \frac{2\pi}{5}, \text{cis } \frac{4\pi}{5}, \text{cis } -\frac{2\pi}{5}, \text{cis } -\frac{4\pi}{5}$



b) Let $\omega = \text{cis } \frac{2\pi}{5}$

$\omega^2 = \text{cis } \frac{4\pi}{5}$

$\omega^3 = \text{cis } \frac{6\pi}{5} = \text{cis } -\frac{4\pi}{5}$

$\omega^4 = \text{cis } \frac{8\pi}{5} = \text{cis } -\frac{2\pi}{5}$

$\therefore 1, \omega, \omega^2, \omega^3, \omega^4$ are the roots of $z^5 - 1 = 0$

$\therefore 1 + \omega + \omega^2 + \omega^3 + \omega^4 = -\frac{b}{a}$

$1 + \omega + \omega^2 + \omega^3 + \omega^4 = 0$

$\therefore \omega + \omega^2 + \omega^3 + \omega^4 = -1$

(2)

Q8 a) Using De Moivre's Theorem

$(\cos \theta + i \sin \theta)^2 = \cos 2\theta + i \sin 2\theta$

But $(\cos \theta + i \sin \theta)^2 = \cos^2 \theta + 2i \sin \theta \cos \theta + i^2 \sin^2 \theta$
 $= \cos^2 \theta - \sin^2 \theta + 2i \sin \theta \cos \theta$

Equating real & imaginary parts

$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$

$\sin 2\theta = 2 \sin \theta \cos \theta$

(2)

b) If $\theta = \frac{\pi}{n}$

$\cos \frac{2\pi}{n} = \cos^2 \frac{\pi}{n} - \sin^2 \frac{\pi}{n}$

$\sin \frac{2\pi}{n} = 2 \sin \frac{\pi}{n} \cos \frac{\pi}{n}$

(1)

c) $\left[1 + \cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n}\right]^n$

$= \left[1 + \cos^2 \frac{\pi}{n} - \sin^2 \frac{\pi}{n} + 2i \sin \frac{\pi}{n} \cos \frac{\pi}{n}\right]^n$

$= \left[1 + \cos^2 \frac{\pi}{n} - (1 - \cos^2 \frac{\pi}{n}) + 2i \sin \frac{\pi}{n} \cos \frac{\pi}{n}\right]^n$

$= \left[2 \cos^2 \frac{\pi}{n} + 2i \sin \frac{\pi}{n} \cos \frac{\pi}{n}\right]^n$

$= \left[2 \cos \frac{\pi}{n} (\cos \frac{\pi}{n} + i \sin \frac{\pi}{n})\right]^n$

$= 2^n \cos^n \frac{\pi}{n} (\cos \frac{\pi}{n} + i \sin \frac{\pi}{n})^n$

(3)

$$= 2^n \cos^n \frac{\pi}{n} \left(\cos n \cdot \frac{\pi}{n} + i \sin n \cdot \frac{\pi}{n} \right) \text{ by D.M.T}$$

$$= 2^n \cos^n \frac{\pi}{n} \left(\cos \pi + i \sin \pi \right)$$

$$= 2^n \cos^n \frac{\pi}{n} (-1 + i \cdot 0)$$

$$= -2^n \cos^n \frac{\pi}{n}$$

(4)

$$29. a) \frac{\pi}{6} + \frac{\pi}{4} = \frac{2\pi + 3\pi}{12}$$

$$= \frac{5\pi}{12}$$

$$\tan \frac{5\pi}{12} = \tan \left(\frac{\pi}{6} + \frac{\pi}{4} \right)$$

$$= \frac{\tan \frac{\pi}{6} + \tan \frac{\pi}{4}}{1 - \tan \frac{\pi}{6} \tan \frac{\pi}{4}}$$

$$= \frac{\frac{1}{\sqrt{3}} + 1}{1 - \frac{1}{\sqrt{3}} \times 1}$$

$$= \frac{\sqrt{3} + 1}{\sqrt{3}}$$

$$\frac{\sqrt{3} - 1}{\sqrt{3}}$$

$$= \frac{\sqrt{3} + 1}{\sqrt{3} - 1}$$

(2)

$$\text{Now } \frac{\pi}{4} - \frac{\pi}{6} = \frac{\pi}{12}$$

$$\therefore \tan \frac{\pi}{12} = \tan \left(\frac{\pi}{4} - \frac{\pi}{6} \right)$$

$$= \frac{\tan \frac{\pi}{4} - \tan \frac{\pi}{6}}{1 + \tan \frac{\pi}{4} \tan \frac{\pi}{6}}$$

$$= \frac{1 - \frac{1}{\sqrt{3}}}{1 + \frac{1}{\sqrt{3}}}$$

$$= \frac{\sqrt{3} - 1}{\sqrt{3} + 1}$$

$$\frac{\sqrt{3} - 1}{\sqrt{3}}$$

$$= \frac{\sqrt{3} - 1}{\sqrt{3} + 1}$$

(2)

$$b) \text{ Let } \sqrt{-6i} = a + ib$$

$$\therefore -6i = (a + ib)^2$$

$$-6i = a^2 - b^2 + 2iab$$

$$\therefore a^2 - b^2 = 0 \Rightarrow 2ab = -6$$

$$a^2 = b^2$$

$$\pm a = \pm b$$

$$\therefore -2a^2 = -6$$

$$a^2 = 3$$

$$a = \pm \sqrt{3}$$

$$b = \mp \sqrt{3}$$

(2)

$$12. \sqrt{-6i} = \sqrt{3} - i\sqrt{3} \text{ or } -\sqrt{3} + i\sqrt{3}$$

$$c) z = \frac{-(1+i) \pm \sqrt{(1+i)^2 - 4 \times 1 \times 2i}}{2}$$

$$= \frac{-(1+i) \pm \sqrt{1+2i+i^2-8i}}{2}$$

$$= \frac{-(1+i) \pm \sqrt{-6i}}{2}$$

$$\text{taking } \sqrt{-6i} \text{ as } \sqrt{3} - i\sqrt{3} \text{ by convention}$$

$$z = \frac{-(1+i) + \sqrt{3} - i\sqrt{3}}{2} \text{ or } \frac{-(1+i) - (\sqrt{3} - i\sqrt{3})}{2}$$

$$= \frac{\sqrt{3}-1}{2} + \frac{(-\sqrt{3}-1)i}{2} \text{ or } \frac{-\sqrt{3}-1}{2} + \frac{(\sqrt{3}-1)i}{2}$$

$$= \frac{(\sqrt{3}-1)}{2} - \frac{(\sqrt{3}+1)i}{2} \text{ or } \frac{-(\sqrt{3}+1)}{2} + \frac{(\sqrt{3}-1)i}{2}$$

$$d) \text{ (i) Let } z_1 = \frac{(\sqrt{3}-1)}{2} - \frac{(\sqrt{3}+1)i}{2}$$

$$\Delta z_2 = \frac{-(\sqrt{3}+1)}{2} + \frac{(\sqrt{3}-1)i}{2}$$

$$|z_1| = \sqrt{\frac{3-2\sqrt{3}+1}{4} + \frac{3+2\sqrt{3}+1}{4}}$$

$$= \sqrt{2}$$

$$|z_2| = \sqrt{\frac{3-2\sqrt{3}+1}{4} + \frac{3-2\sqrt{3}+1}{4}}$$

$$= \sqrt{2}$$

(1)

$$\text{ii) } \arg(z_1) = \tan^{-1} \left[\frac{-\left(\frac{\sqrt{3}+1}{2}\right)}{\frac{(\sqrt{3}-1)}{2}} \right]$$

$$= -\tan^{-1} \left[\frac{\sqrt{3}+1}{\sqrt{3}-1} \right]$$

$$= -\frac{5\pi}{12}$$

$$\arg(z_2) = \pi - \tan^{-1} \left[\frac{\frac{(\sqrt{3}-1)}{2}}{\frac{(\sqrt{3}+1)}{2}} \right]$$

$$= \pi - \tan^{-1} \left[\frac{\sqrt{3}-1}{\sqrt{3}+1} \right]$$

$$= \pi - \frac{\pi}{12}$$

$$= \frac{11\pi}{12}$$

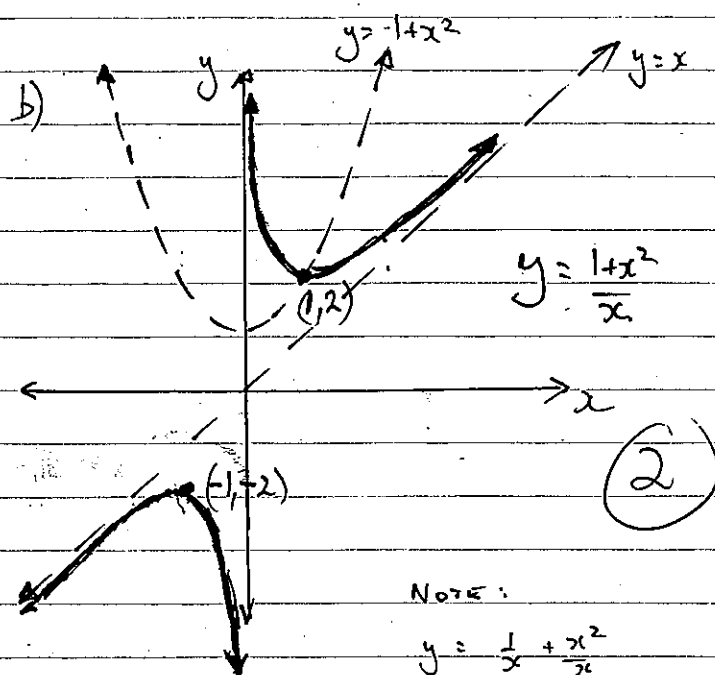
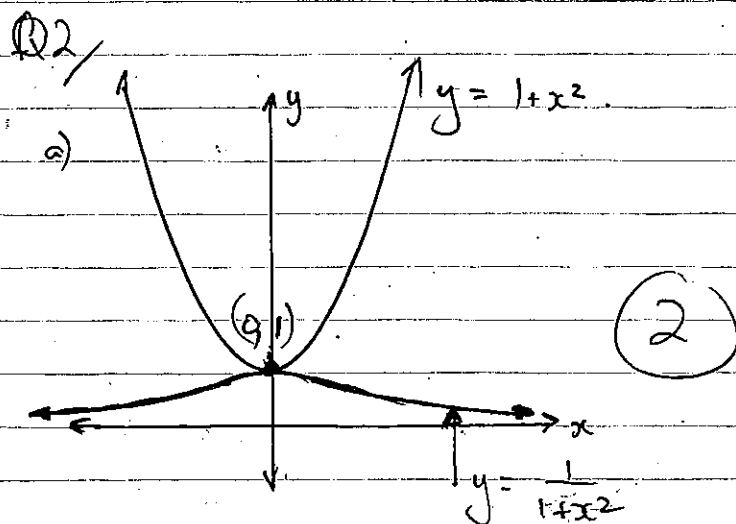
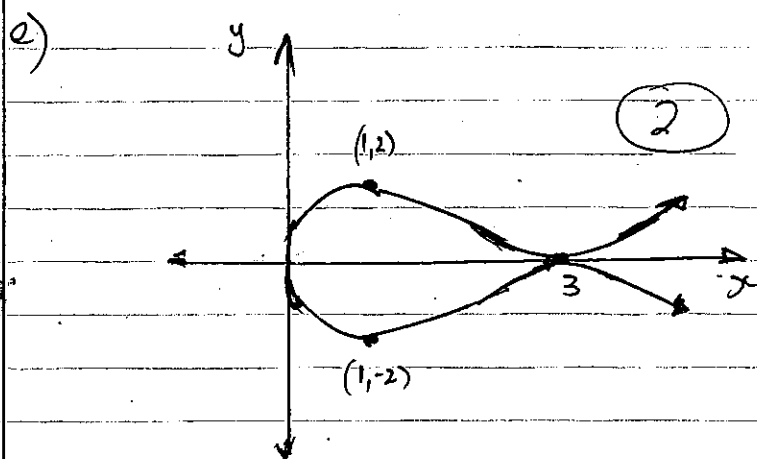
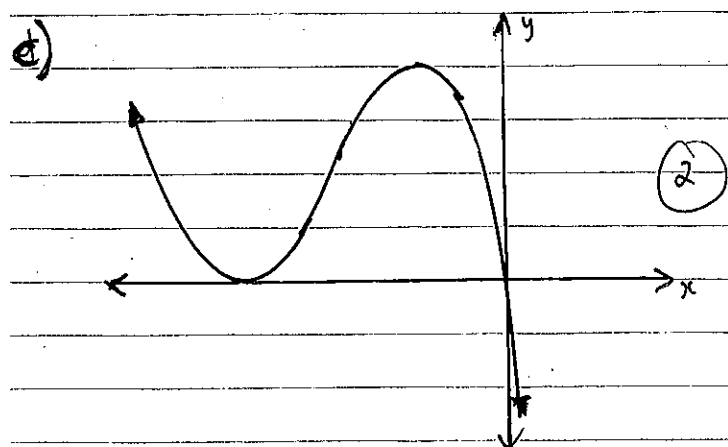
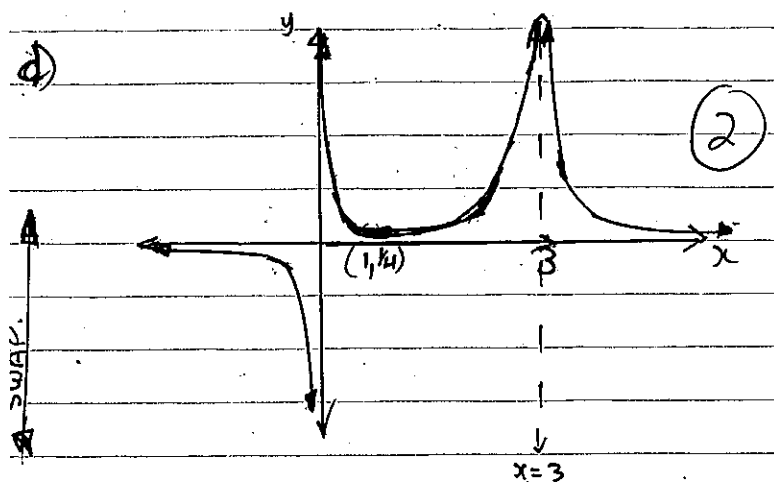
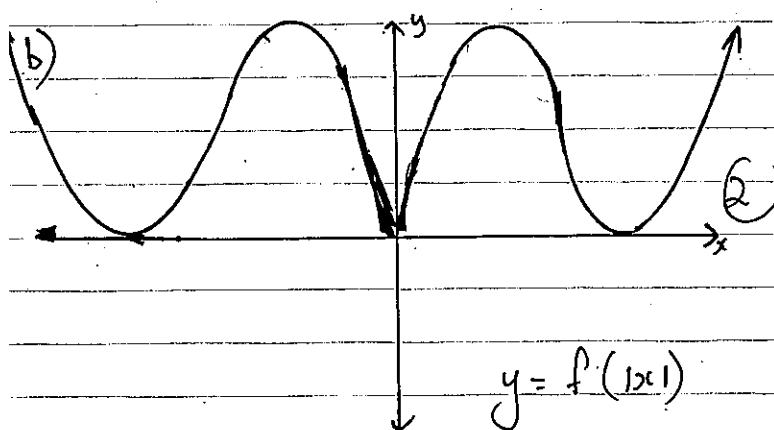
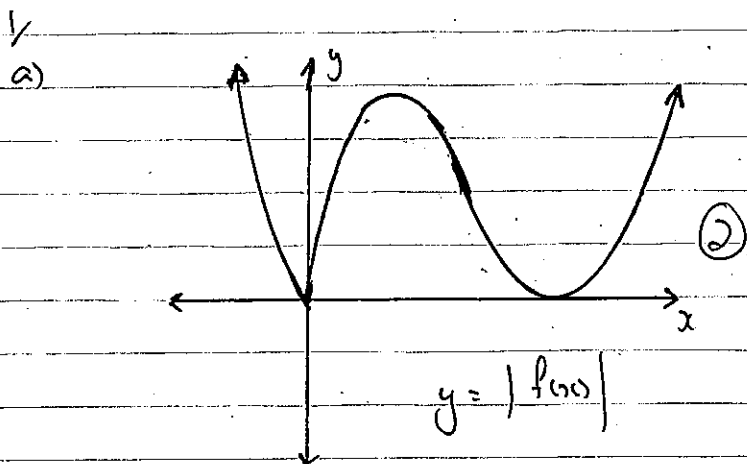
(2)

$$\therefore \arg(z_1) + \arg(z_2) = -\frac{5\pi}{12} + \frac{11\pi}{12}$$

$$= \frac{\pi}{2}$$

SOLUTIONS.

GRAPHS



NOTE:

$$y = \frac{1}{x} + \frac{x^2}{x}$$

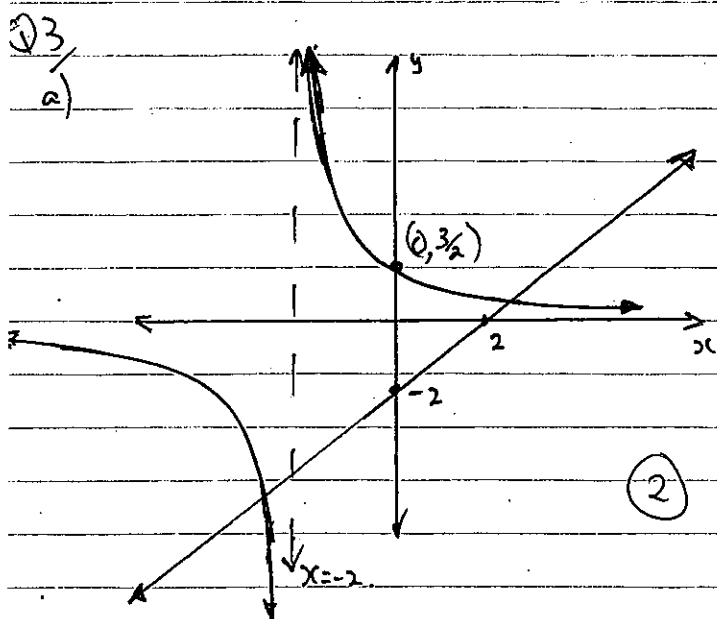
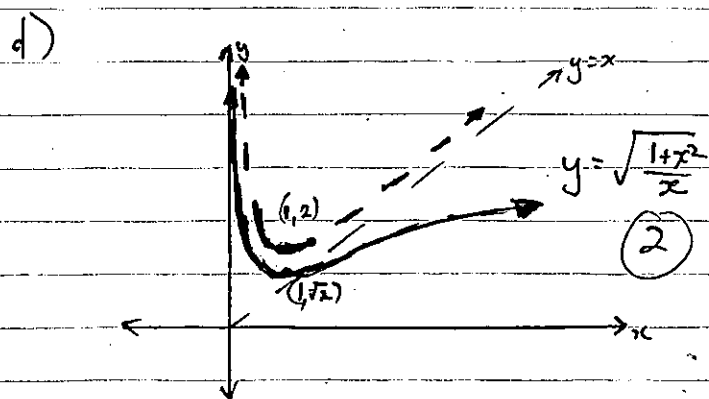
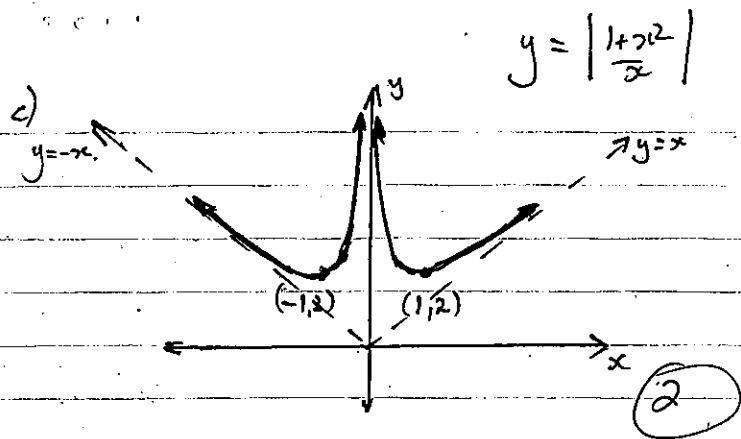
$$= \frac{1}{x} + x$$

$$y' = -\frac{1}{x^2} + 1$$

$$= \frac{-1}{x^2} + 1$$

If $y' = 0$, $\frac{1}{x^2} = 1$, $\therefore x = \pm 1$

$y = \pm 2$



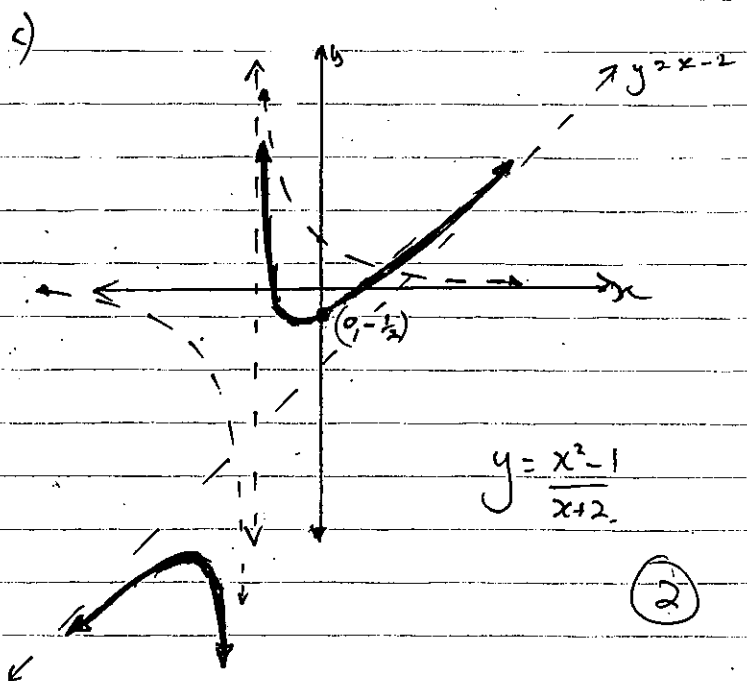
b) RHS = $x-2 + \frac{3}{x+2}$

$$= \frac{(x-2)(x+2) + 3}{x+2}$$

$$= \frac{x^2 - 4 + 3}{x+2}$$

$$= \frac{x^2 - 1}{x+2}$$

②



d) If $\frac{x^2-1}{x+2} = 0$

$$x^2 - 1 = 0$$

$$x = \pm 1$$

$\therefore \frac{x^2-1}{x+2} \leq 0$ when

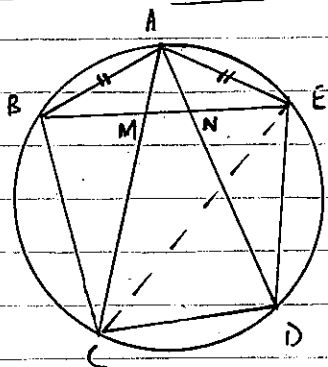
②

$$x < -2 \text{ OR } -1 \leq x \leq 1$$

SOLUTIONS.

CIRCLE GEOMETRY

Q1.



a) Join C to E

$\angle BEA = \angle ABE$ (base $\angle$'s of an isosc Δ are equal, $AB = AE$)

$\angle ABE = \angle ACE$ ($\angle$'s in the same segment standing on AE are equal)

$\therefore \angle BEA = \angle ACE$ (2)

b) $\angle EAD = \angle ECD$ ($\angle$'s in the same segment standing on ED are equal)

$\therefore \angle BEA + \angle EAD = \angle ACE + \angle ECD$

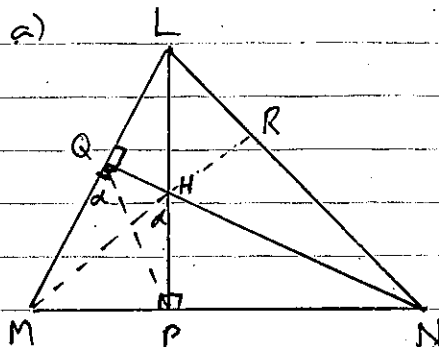
But $\angle BEA + \angle EAD = \angle END$ (ext $\angle$ of a Δ theorem)

$\therefore \angle ACE + \angle ECD = \angle ACD$

$\therefore \angle END = \angle ACD$

Hence, C, D, N, M is a cyclic quad as the ext $\angle$ is equal to the opp. int. $\angle$. (3)

Q2 a)



(2)

a) Join M to H & P to Q

Now PQMH is a cyclic quadrilateral as $\angle$'s MQH & MPH are opp. suppl. $\angle$'s.

$\therefore \angle PHM = \angle PQM$ ($\angle$'s in the same segment standing on MP are equal). (2)

b) Now $\angle QPN$ is a cyclic quad as $\angle QPN$ & $\angle LPN$ form a pair of equal $\angle$'s in the same segment on arc LN.

Let $\angle PHM = \angle PQM$ be α

$\angle LQP = 180^\circ - \alpha$ (adj. suppl. $\angle$'s)

$\therefore \angle LNP = \alpha$ (opp $\angle$'s of a cyclic quad are suppl.)

$\therefore \angle PHM = \angle LNM$ (both α) (2)

c) $\angle RHP = 180^\circ - \alpha$ (adj. suppl. $\angle$'s)

$\therefore \angle RHP$ & $\angle RNP$ form a pair of opp. suppl. $\angle$'s

$\therefore RHPN$ is cyclic.

$\therefore \angle NRH + \angle NPH = 180^\circ$ (opp $\angle$'s of a cyclic quad)

i.e. $\angle NRH + 90 = 180$

$\therefore MR \perp LN$. (3)